6. Weight of the water drop = 0.018 g

No. of moles of water in the drop = Mass of water / molar mass

= 0.018 / 18 = 10^{-3} mole

No of water molecules present in 1 mole of water = 6.022×10^{23}

No. water molecules in one drop of water (10^{-3} mole) = $6.022 \times 10^{23} \times 10^{-3}$

= **6.022×10^{20}**

7. $MgCO_3 \rightarrow MgO + CO_2 \uparrow$

$MgCO_3$: (1 x 24) + (1 x 12) + (3 x 16) = 84 g

CO_2 : (1 x 12) + (2 x 16) = 44 g

100 % pure 84 g $MgCO_3$ on heating gives 44 g CO_2

Given that 1 g of $MgCO_3$ on heating gives 0.44 g CO_2

Therefore, 84 g $MgCO_3$ sample on heating gives 36.96 g CO_2

$$\text{Percentage of purity of the sample} = \frac{100\,\%}{44\ \text{g}\ CO_2} \times 36.96\ \text{g}\ CO_2$$
$$= 84\,\%$$

Percentage of impurity = **16 %**

8. $NaHCO_3 + CH_3COOH \rightarrow \underbrace{CH_3COONa + H_2O}\ + CO_2 \uparrow$

6.3 g + 30 g → 33 g + x

The amount of CO_2 released, x = 3.3 g

No. of moles of CO_2 released = 3.3 / 44 = 0.075 mol

9 $H_2(g) + Cl_2(g) \rightarrow 2\ HCl\ (g)$

Content	**$H_2(g)$**	**$Cl_2(g)$**	**HCl (g)**
Stoichiometric coefficient	1	1	2
No. of moles of reactants allowed to react at 273 K and 1 atm pressure	22.4 L (1 mol)	11.2 L (0.5 mol)	-
No. of moles of reactant reacted and product formed	0.5	0.5	1
Amount of HCl formed = 1 mol			

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